COURSE TITLE	**BUSINESS STATISTICS & ECONOMETRICS**
COURSE CODE	**04MI0105**
COURSE CREDITS	**3**

Course Outcomes: After completion of this course, student will be able to:

1. Apply descriptive statistical measures and basic probability concepts to summarize and interpret data.
2. Analyze real-life situations using appropriate probability distributions and compute expected values.
3. Demonstrate the ability to draw valid inferences about populations using sampling techniques and estimation methods
4. Conduct hypothesis testing and interpret correlation and regression outputs in business and economic contexts
5. Estimate and interpret simple econometric models using Ordinary Least Squares and recognize basic violations of model assumptions

Pre-requisite of course: Basic knowledge of mathematics, statistics, and introductory economics.

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	0	0	50	30	20	0	0

Contents : Unit	Topics	Contact Hours
1	**Descriptive Statistics and Probability** Concepts of statistics and probability, Types of data: qualitative and quantitative, Data organization and summarization, Measures of central tendency: mean, median, mode, Measures of dispersion: range, variance, standard deviation, coefficient of variation, Basic probability rules, Conditional probability, Bayes' theorem	9
2	**Random Variables and Probability Distributions** Random variables: discrete and continuous, Probability distributions: Binomial, Poisson, Normal, Properties and applications of theoretical distributions, Expected value, Variance of random variables, Quantitative analysis of uncertainty	9
3	**Sampling and Estimation** Population and sample concepts, Sampling methods: random, stratified, systematic, Sampling distribution of sample mean and proportion, Central Limit Theorem, Point estimation, Interval estimation, Confidence intervals for population mean and proportion	9

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Contents : Unit	Topics	Contact Hours
4	**Hypothesis Testing and Correlation Analysis** Formulation of null and alternative hypotheses, Type I and Type II errors, , p-values and significance testing, Z-test, t-test (one and two samples), Chi-square test, Pearson's correlation coefficient, Simple linear regression, Interpretation of regression coefficients	9
5	**Introduction to Econometrics** Definition and importance of econometrics, Classical Linear Regression Model (CLRM), Ordinary Least Squares (OLS) method, Assumptions of CLRM, Interpretation of regression results, Multicollinearity, Heteroscedasticity, Autocorrelation, Consequences and basic remedies	9
Total Hours		**45**

Textbook :

1 Fundamentals of mathematical statistics, Gupta, S. C., & Kapoor, V. K, Sultan Chand & Sons, 2019

2 Introduction to econometrics , Maddala, G. S., & Lahiri, K, Wiley India Edition, 2009

3 Business statistics, Sharma, J. K, Pearson Education India., -

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery and evaluation

Remember / Knowledge	Understand	Apply	Analyze	Evaluate	Higher order Thinking / Creative
10.00	20.00	20.00	20.00	20.00	10.00

Instructional Method:

1 Lectures, practical problem-solving, data analysis using software, and case-based discussions.

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